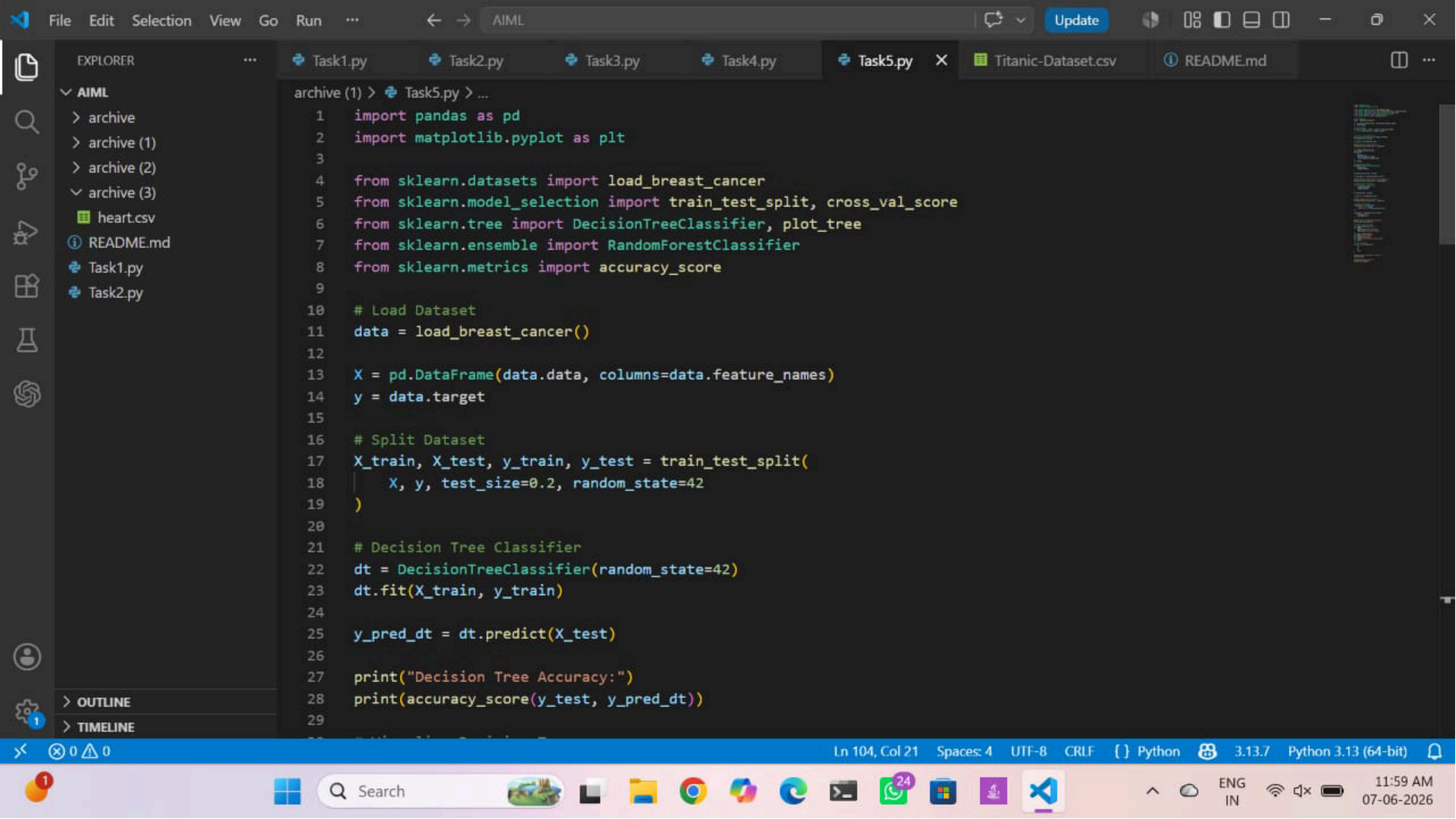
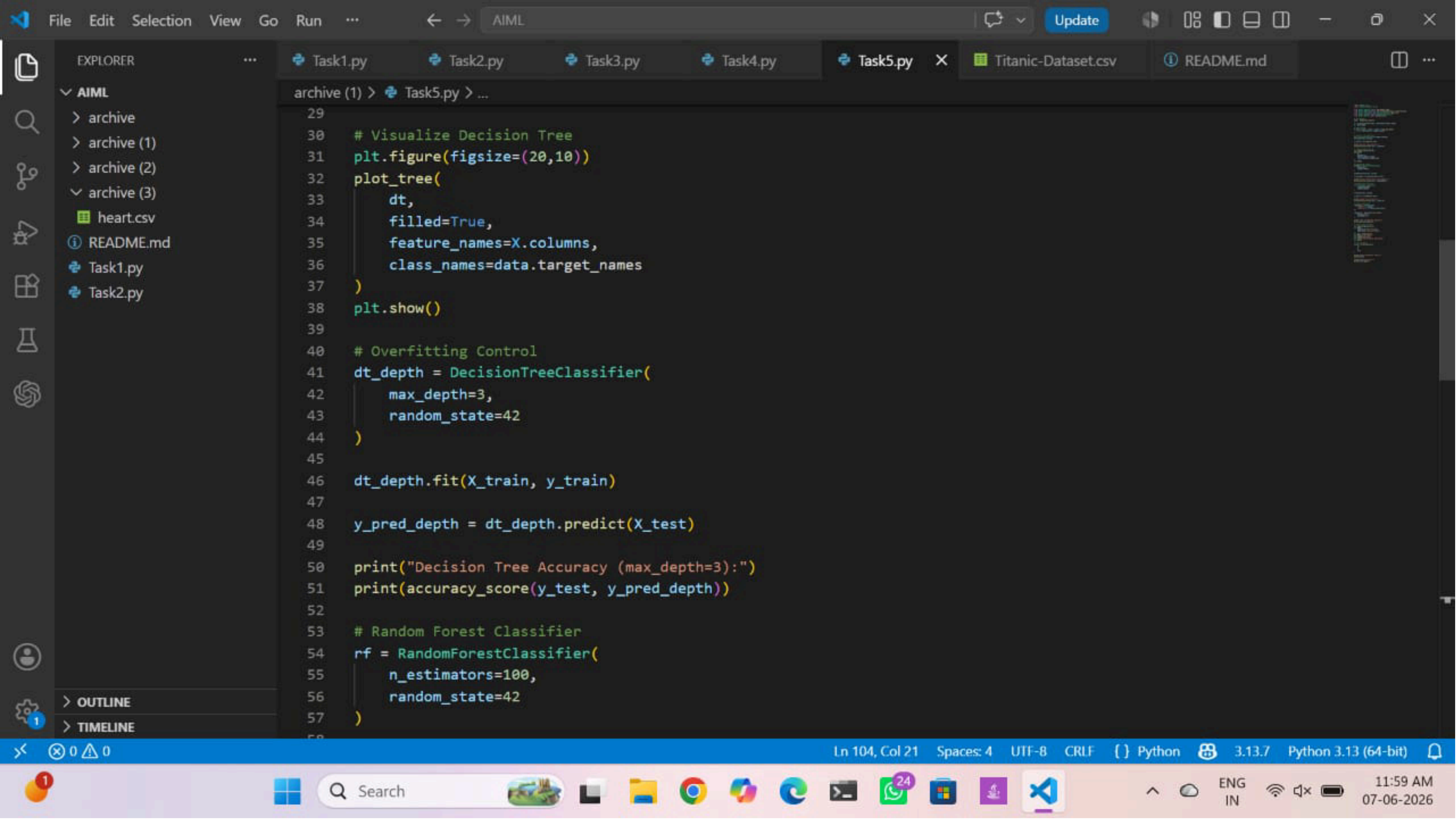
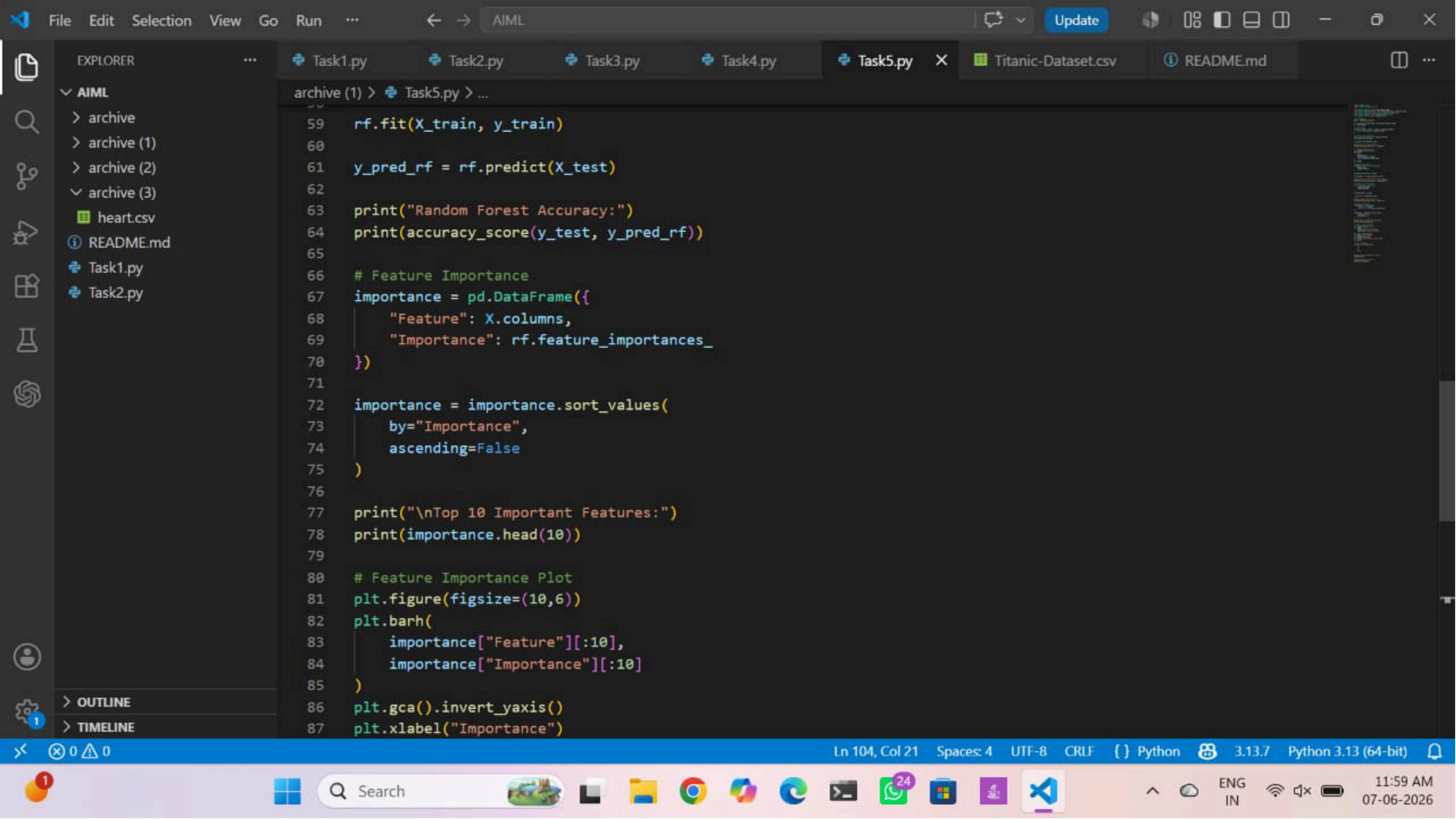


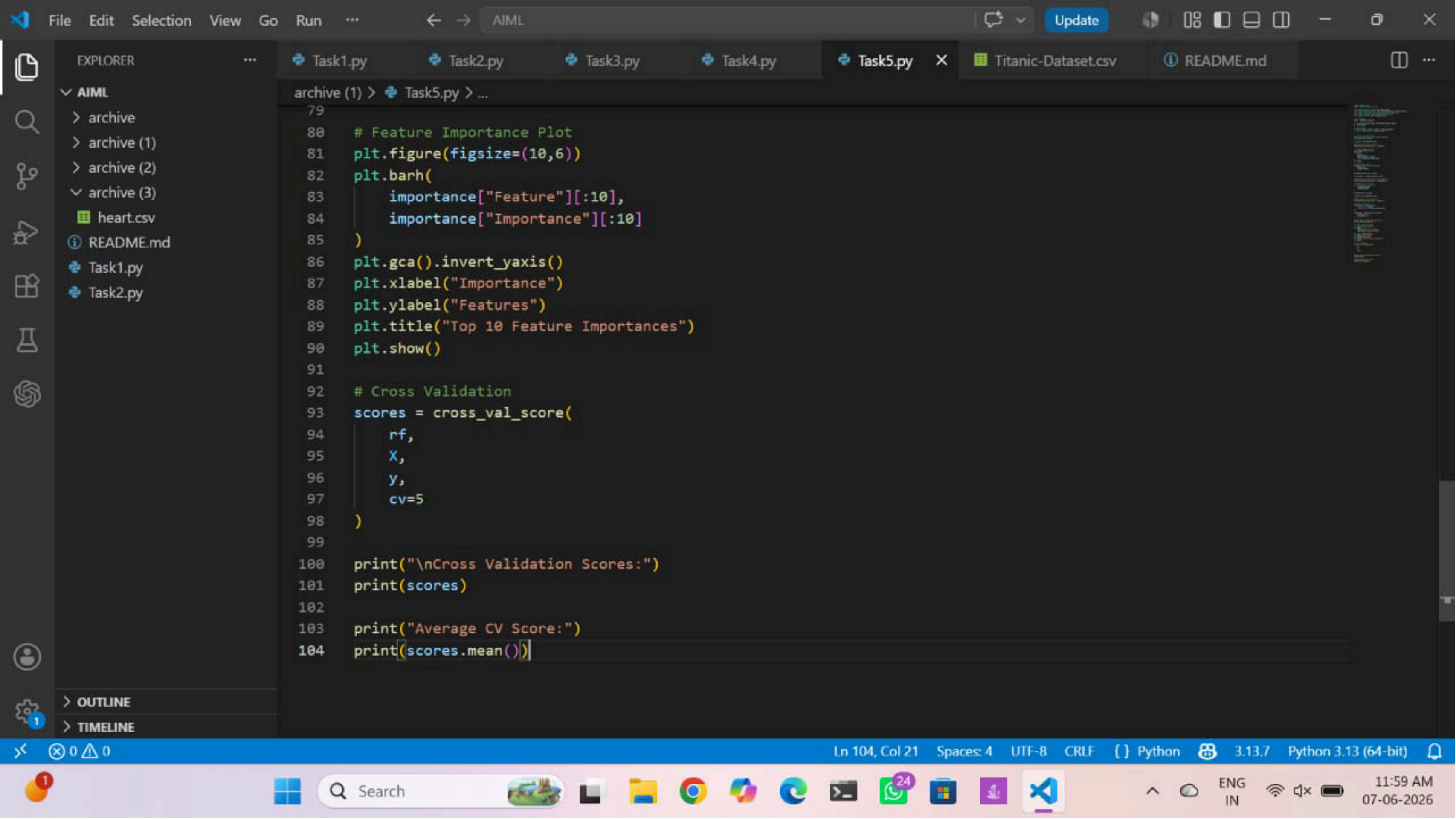




```
File Edit Selection View Go Run ... archive (1) Task5.py Titanic-Dataset.csv README.md
EXPLORER
AIML
  archive
  archive (1)
  archive (2)
  archive (3)
  heart.csv
  README.md
  Task1.py
  Task2.py
  archive (1) > Task5.py > ...
29
30 # Visualize Decision Tree
31 plt.figure(figsize=(20,10))
32 plot_tree(
33     dt,
34     filled=True,
35     feature_names=X.columns,
36     class_names=data.target_names
37 )
38 plt.show()
39
40 # Overfitting Control
41 dt_depth = DecisionTreeClassifier(
42     max_depth=3,
43     random_state=42
44 )
45
46 dt_depth.fit(X_train, y_train)
47
48 y_pred_depth = dt_depth.predict(X_test)
49
50 print("Decision Tree Accuracy (max_depth=3):")
51 print(accuracy_score(y_test, y_pred_depth))
52
53 # Random Forest Classifier
54 rf = RandomForestClassifier(
55     n_estimators=100,
56     random_state=42
57 )
58
```

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File Edit Selection View Go Run ... archive (1) Task1.py Task2.py Task3.py Task4.py Task5.py Titanic-Dataset.csv README.md
AIML
  archive
  archive (1)
  archive (2)
  archive (3)
    heart.csv
    README.md
    Task1.py
    Task2.py
archive (1) > Task5.py > ...
59 rf.fit(X_train, y_train)
60
61 y_pred_rf = rf.predict(X_test)
62
63 print("Random Forest Accuracy:")
64 print(accuracy_score(y_test, y_pred_rf))
65
66 # Feature Importance
67 importance = pd.DataFrame({
68     "Feature": X.columns,
69     "Importance": rf.feature_importances_
70 })
71
72 importance = importance.sort_values(
73     by="Importance",
74     ascending=False
75 )
76
77 print("\nTop 10 Important Features:")
78 print(importance.head(10))
79
80 # Feature Importance Plot
81 plt.figure(figsize=(10,6))
82 plt.barh(
83     importance["Feature"][:10],
84     importance["Importance"][:10]
85 )
86 plt.gca().invert_yaxis()
87 plt.xlabel("Importance")
```



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EXPLORER
AIML
  archive
  archive (1)
  archive (2)
  archive (3)
  heart.csv
  README.md
  Task1.py
  Task2.py
  Task3.py
  Task4.py
  Task5.py
  Titanic-Dataset.csv
  README.md
archive (1) > Task5.py > ...
79
80 # Feature Importance Plot
81 plt.figure(figsize=(10,6))
82 plt.barh(
83     importance["Feature"][:10],
84     importance["Importance"][:10]
85 )
86 plt.gca().invert_yaxis()
87 plt.xlabel("Importance")
88 plt.ylabel("Features")
89 plt.title("Top 10 Feature Importances")
90 plt.show()
91
92 # Cross Validation
93 scores = cross_val_score(
94     rf,
95     X,
96     y,
97     cv=5
98 )
99
100 print("\nCross Validation Scores:")
101 print(scores)
102
103 print("Average CV Score:")
104 print(scores.mean())
```